Bird Species Observation Analysis Report

Introduction

Bird monitoring systems generate large volumes of data related to bird species, their habitats, and surrounding environmental conditions. Analyzing this data helps in understanding patterns in bird distribution, species diversity, and the influence of environmental factors on bird activity.

This project focuses on analyzing bird observation data collected from two different habitats: forest and grassland. The dataset contains records from multiple locations, with each observation including details such as species name, date, location, and weather conditions. Python is used for data cleaning and exploratory data analysis, while MySQL is used for structured querying and analysis. An interactive Power BI dashboard is developed to visualize key insights and trends in bird activity.

The goal of this project is to identify patterns in bird behavior, compare habitat-wise distribution, and understand how environmental conditions such as temperature and weather impact bird observations. The insights generated from this analysis can support biodiversity studies and environmental planning.

Problem Statement

- Bird observation data is distributed across multiple Excel files and sheets, making it difficult to analyze efficiently.
- There is no clear understanding of how bird activity differs between forest and grassland habitats.
- It is challenging to identify which bird species are most common or rare across different locations.
- The impact of environmental factors such as temperature, sky condition, and wind on bird observations is not clearly understood.
- Lack of a structured and cleaned dataset limits the ability to perform meaningful analysis.
- There is no centralized system to generate insights about bird distribution and biodiversity.
- The objective is to analyze bird observation data to uncover patterns in species distribution, habitat preference, and environmental influence.

Objectives

The main objectives of this project are:

- Understand bird distribution across forest and grassland habitats.
- Identify the most common and diverse bird species.
- Analyze bird activity patterns over time (monthly trends).

- Examine the impact of environmental factors such as temperature and weather on bird observations.
- Study location-wise bird activity and biodiversity.
- Create visualizations and dashboards for better data interpretation.
- Generate insights that support ecological research and conservation planning.

Dataset Description

The dataset used in this project contains bird observation records collected from two different habitats: forest and grassland. The data is stored in two Excel files, each consisting of multiple sheets representing different locations such as ANTI, CATO, CHOH, and others. Each sheet contains observations recorded at a specific site.

Each row in the dataset represents a single bird observation and includes details about the species, location, date, and environmental conditions at the time of observation.

Key Columns Description

- **admin_unit_code**: Code representing the administrative unit or location
- **site_name**: Name of the observation site
- **plot_name**: Specific plot or area within the site
- **location_type**: Type of habitat (Forest or Grassland)
- **year**: Year when the observation was recorded
- **date**: Date of observation
- **observer**: Name of the person who recorded the observation
- **visit**: Visit number for the observation
- **id_method**: Method used for identifying the bird species
- **flyover_observed**: Indicates whether the bird was flying over
- **sex**: Gender of the observed bird (if available)
- **common_name**: Common name of the bird species
- **scientific_name**: Scientific name of the bird species
- **acceptedtsn**: Taxonomic Serial Number for species identification
- **npstaxoncode**: Unique taxon code assigned to species
- **aou_code**: American Ornithological Union code for species
- **pif_watchlist_status**: Indicates conservation priority status
- **regional_stewardship_status**: Indicates regional conservation importance
- **temperature**: Temperature recorded during observation
- **humidity**: Humidity level during observation
- **sky**: Sky/weather condition

- **wind**: Wind condition
- **disturbance**: Level of disturbance in the area
- **initial_three_min_cnt**: Indicates if the bird was observed in the initial 3 minutes
- **sheet**: Represents the location (derived from sheet name)
- **previously_obs**: Indicates whether the species was previously observed
- **dataset**: Type of dataset (Forest or Grassland)
- **month**: Month derived from the observation date
- **temp_bin**: Categorized temperature ranges for analysis

The final combined dataset contains 17,077 records and 27 columns, providing a comprehensive view of bird observations across different habitats and environmental conditions.

Tools Used

The following tools were used in this project:

- **Python**: Data cleaning and exploratory data analysis
- **SQL (MySQL)**: Querying data and calculating KPIs
- **Power Bi** : Dashboard creation and data visualization

Python EDA Summary

Exploratory Data Analysis (EDA) was performed using Python to understand, clean, and transform the bird observation dataset before conducting further analysis and visualization.

Data Preprocessing & Feature Engineering

During the EDA process, additional columns were created to support better analysis:

- **month**: Extracted from the date column to analyze bird activity across different months.
- **dataset**: Added to distinguish between forest and grassland data.
- **temp_bin**: Created by grouping temperature values into ranges to study the effect of temperature on bird activity.

These features helped in identifying temporal patterns and environmental impacts on bird observations.

Handling Missing Values

- The **sub_unit_code** column contained a large number of missing values and was removed as it was not useful for analysis.
- Missing values in **sex** were filled with "Unknown" to maintain consistency.
- Columns such as **acceptedtsn** and **id_method** had minimal missing values and were handled appropriately using forward fill techniques.

- Columns with completely missing values (e.g., interval_length, distance in some cases) were removed to improve data quality.

EDA Findings Summary

- Bird observations are higher in forest habitats compared to grassland.
- Certain bird species appear more frequently, indicating dominant species in the dataset.
- Bird activity varies across different months, showing seasonal patterns.
- Environmental factors such as temperature and sky conditions influence bird observations.
- Some locations (sheets) show higher bird activity and biodiversity than others.
- Most observations are recorded as non-flyover, indicating birds are often observed within the habitat rather than flying over.

Overall, Python EDA helped in cleaning the dataset, creating useful features, and identifying key patterns that were further explored using SQL and Power BI.

SQL Queries & Analysis

SQL Queries & Analysis

SQL was used to perform structured analysis on the cleaned bird observation dataset stored in MySQL. These queries helped validate insights obtained from Python EDA and provided deeper understanding of bird distribution, species diversity, habitat preference, and environmental impact.

1. Total Bird Observations

Objective:
To determine the total number of bird observations in the dataset.

```
SELECT COUNT(*) AS total_observations

FROM bird;
```

Analysis:
This query provides the total number of observations, helping understand the dataset size.

2. Habitat Comparison (Forest vs Grassland)

Objective:
To compare bird activity between forest and grassland habitats.

```
SELECT dataset, COUNT(*) AS total_birds

FROM bird

GROUP BY dataset;
```

Analysis:
This shows which habitat has more bird observations.

3. Unique Species Count (Biodiversity)

Objective:
To measure biodiversity across habitats.

```
SELECT dataset, COUNT(DISTINCT common_name) AS unique_species

FROM bird

GROUP BY dataset;
```

Analysis:
This identifies which habitat has greater species diversity.

4. Top 10 Bird Species

Objective:
To identify the most frequently observed bird species.

```
SELECT common_name, COUNT(*) AS count

FROM bird

GROUP BY common_name

ORDER BY count DESC

LIMIT 10;
```

Analysis:
This highlights the most common bird species.

5. Species Preference (Habitat-wise)

Objective:
To analyze species distribution across habitats.

```
SELECT dataset, common_name, COUNT(*) AS count

FROM bird

GROUP BY dataset, common_name

ORDER BY count DESC;
```

Analysis:
This shows which species prefer forest or grassland.

6. Year-wise Analysis

Objective:
To analyze bird observations by year.

```
SELECT year, dataset, COUNT(*) AS total

FROM bird

GROUP BY year, dataset

ORDER BY year;
```

Analysis:
Since the dataset contains only one year, this confirms data consistency.

7. Monthly Trend

Objective:
To analyze bird activity across months.

```
SELECT month, dataset, COUNT(*) AS total

FROM bird

GROUP BY month, dataset

ORDER BY month;
```

Analysis:
This helps identify seasonal patterns in bird observations.

8. Location-wise Analysis

Objective:
To analyze bird observations across different locations.

```
SELECT sheet, COUNT(*) AS total_birds

FROM bird

GROUP BY sheet

ORDER BY total_birds DESC;
```

Analysis:
This identifies locations with the highest bird activity.

9. Top Locations by Biodiversity

Objective:
To identify locations with highest species diversity.

```
SELECT sheet, COUNT(DISTINCT common_name) AS species_count

FROM bird

GROUP BY sheet

ORDER BY species_count DESC;
```

Analysis:
This shows which locations have the most diverse bird species.

10. Weather Impact (Sky Condition)

Objective:
To analyze how sky conditions affect bird observations.

```
SELECT sky, COUNT(*) AS total
```

```
FROM bird

GROUP BY sky

ORDER BY total DESC;
```

Analysis:
This shows how weather influences bird activity.

11. Temperature Impact

Objective:
To study bird activity across temperature ranges.

```
SELECT temp_bin, COUNT(*) AS total

FROM bird

GROUP BY temp_bin

ORDER BY total DESC;
```

Analysis:
This helps understand the effect of temperature on bird observations.

12. Wind Effect

Objective:
To analyze the effect of wind conditions.

```
SELECT wind, COUNT(*) AS total

FROM bird

GROUP BY wind

ORDER BY total DESC;
```

Analysis:
This shows how wind conditions impact bird observations.

13. Observer Contribution

Objective:
To identify top contributors to data collection.

```
SELECT observer, COUNT(*) AS total

FROM bird

GROUP BY observer

ORDER BY total DESC

LIMIT 10;
```

Analysis:
This highlights the most active observers.

14. Flyover vs Non-Flyover Observations

Objective:
To analyze bird observation type.

```
SELECT flyover_observed, COUNT(*) AS total

FROM bird

GROUP BY flyover_observed;
```

Analysis:
This shows whether birds are mostly observed flying or within habitat.

15. Initial 3-Minute Count Analysis

Objective:
To analyze early observation counts.

```
SELECT initial_three_min_cnt, COUNT(*) AS total

FROM bird

GROUP BY initial_three_min_cnt;
```

Analysis:
This helps understand early detection patterns.

16. Temperature vs Habitat Analysis

Objective:
To compare bird activity across temperature ranges and habitats.

```
SELECT temp_bin, dataset, COUNT(*) AS total

FROM bird

GROUP BY temp_bin, dataset

ORDER BY total DESC;
```

Analysis:
This shows how temperature affects bird activity in different habitats.

17. Habitat + Weather Combined

Objective:
To analyze combined effect of habitat and weather.

```
SELECT dataset, sky, COUNT(*) AS total

FROM bird

GROUP BY dataset, sky

ORDER BY total DESC;
```

Analysis:
This provides deeper insights into environmental influence.

18. Most Common Bird per Habitat

Objective:
To identify dominant species in each habitat.

```
SELECT dataset, common_name, COUNT(*) AS count

FROM bird

GROUP BY dataset, common_name

ORDER BY dataset, count DESC;
```

Analysis:
This shows top species in forest and grassland.

19. Rare Species

Objective:
To identify less frequently observed species.

```
SELECT common_name, COUNT(*) AS count

FROM bird

GROUP BY common_name

HAVING count < 5;
```

Analysis:
This helps identify rare species in the dataset.

SQL analysis provided structured insights into bird distribution, biodiversity, habitat preference, and environmental impact, supporting the findings from Python EDA and enhancing the overall analysis.

Dashboard Overview

An interactive Power BI dashboard was developed to visualize bird observation data and provide meaningful insights into species distribution, habitat comparison, and environmental impact. The dashboard is designed to be user-friendly and allows dynamic exploration through filters and visuals.

1. Dashboard KPIs

The top section of the dashboard displays key performance indicators (KPIs) that summarize overall bird observation data:

- **Total Birds**: Total number of bird observations (~17K)
- **Unique Species**: Total number of distinct bird species (126)
- **Average Temperature**: Average temperature recorded during observations (~22.57°C)

These KPIs provide a quick overview of the dataset and overall bird activity.

2. Total Birds by Habitat

This chart compares bird observations between forest and grassland habitats:

- Forest and grassland show comparable bird counts
- Helps understand habitat-wise bird distribution
- Supports comparison of bird activity across environments

3. Total Birds by Species

This bar chart highlights the most commonly observed bird species:

- Shows top bird species such as Northern Cardinal and Carolina Wren
- Helps identify dominant species in the dataset
- Useful for biodiversity analysis

4. Total Birds by Visit

This chart shows bird observations across different visit numbers:

- Indicates variation in observations across visits
- Helps understand how observation frequency changes over time

5. Total Birds by Sheet (Location Analysis)

This visualization displays bird observations across different locations:

- Certain locations (e.g., ANTI, MONO) show higher bird activity
- Helps identify high-density bird observation areas
- Useful for location-based biodiversity analysis

6. Total Birds by Temperature

This chart shows bird activity across different temperature ranges:

- Moderate temperature ranges have higher bird observations
- Extreme temperature ranges show lower activity
- Helps understand environmental impact on bird behavior

7. Total Birds by Sky Condition

This chart analyzes bird observations based on weather conditions:

- Partly cloudy and clear conditions show higher bird activity
- Fog and drizzle conditions have fewer observations
- Highlights the influence of weather on bird visibility and activity

8. Interactive Filters (Slicers)

The dashboard includes interactive filters that allow users to explore data dynamically:

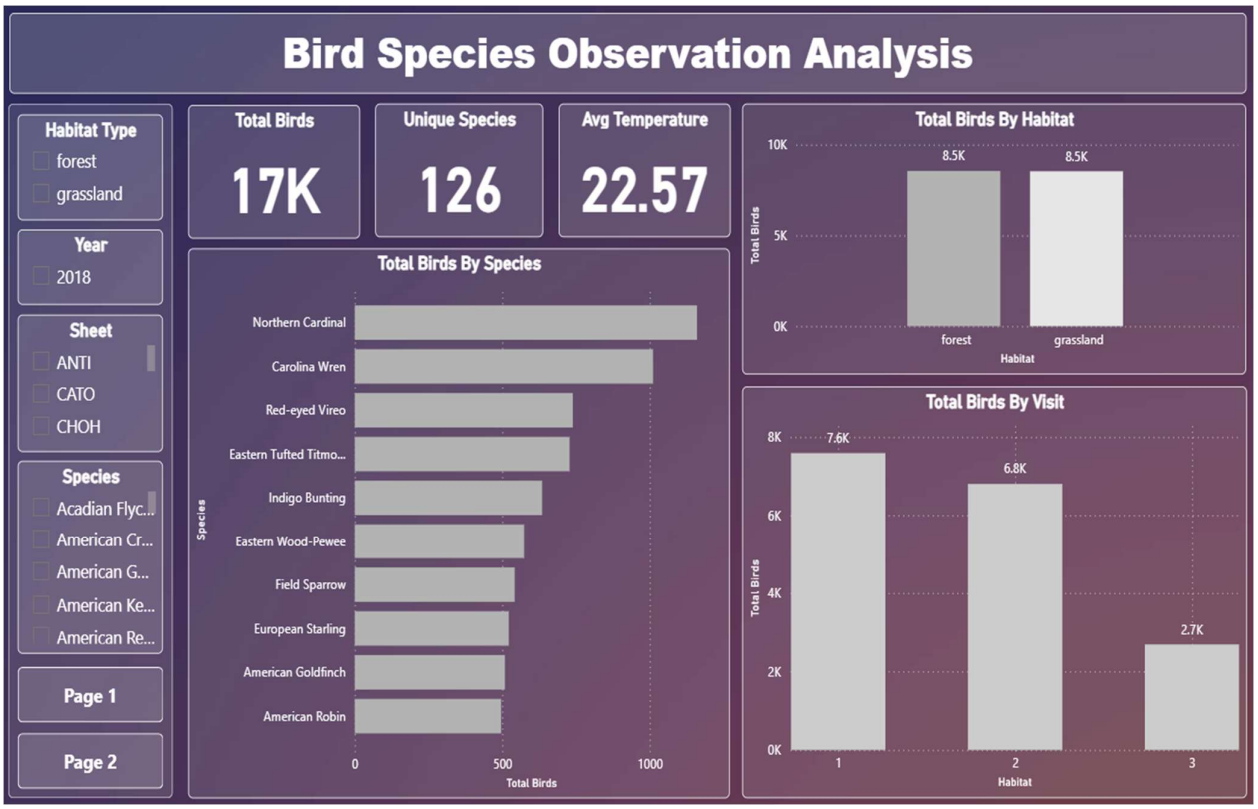
- **Habitat Type**: Filter by forest or grassland
- **Year**: Filter observations by year
- **Sheet (Location)**: Select specific locations
- **Species**: Analyze specific bird species

These filters enhance user interaction and allow detailed analysis of specific segments.

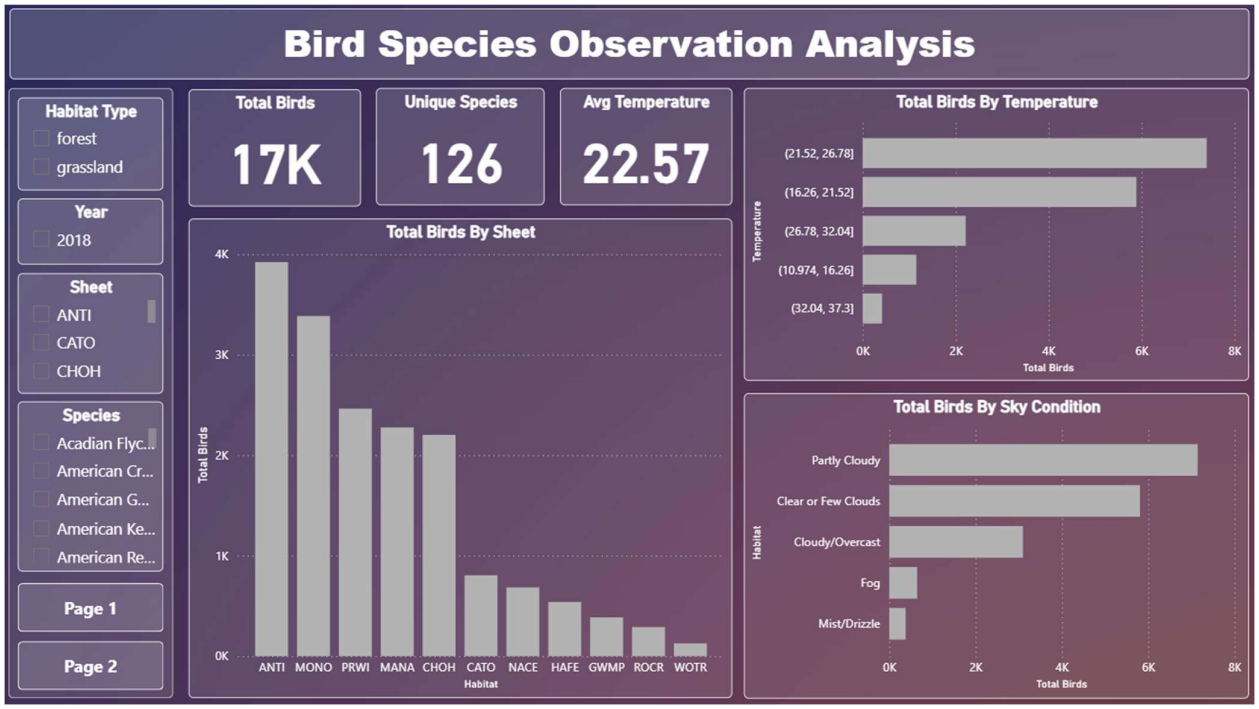
Key Insights

- Bird observations are slightly higher in forest habitats compared to grassland, indicating richer bird activity in forest areas.
- A small number of species dominate the dataset, with species like Northern Cardinal and Carolina Wren being the most frequently observed.
- Certain locations such as ANTI and MONO show significantly higher bird observations, indicating biodiversity hotspots.
- Bird activity is higher in moderate temperature ranges, suggesting favorable environmental conditions.
- Weather conditions like partly cloudy and clear skies result in higher bird observations, while fog and drizzle reduce activity.
- Observation counts vary across visits, indicating differences in bird detection or survey patterns.

Page 1



Page 2



Recommendations

Based on the analysis performed using Python, SQL, and Power BI, the following recommendations are suggested:

- **Focus on high biodiversity locations**
  Locations such as ANTI and MONO show higher bird activity and species diversity. These areas should be prioritized for conservation and monitoring efforts.

- **Enhance habitat conservation in forests**
  Forest habitats show slightly higher bird observations, indicating richer biodiversity. Protecting forest ecosystems can help maintain and improve bird populations.

- **Improve monitoring in grassland areas**
  Since grassland shows comparatively lower observations, increasing monitoring efforts can help better understand species presence and habitat conditions.

- **Consider environmental factors in planning**
  Bird activity is higher under moderate temperatures and favorable sky conditions. Environmental factors should be considered when planning observation schedules and conservation strategies.

- **Promote consistent observation practices**
  Variation in observations across visits suggests inconsistency in data collection. Standardizing observation methods can improve data quality and reliability.

- **Focus on rare species conservation**
  Species with very low observation counts should be monitored closely to ensure they are not at risk and to support biodiversity preservation.

- **Leverage data-driven decision making**
  Insights from data analysis should be used by environmental organizations and policymakers to design better conservation strategies and habitat management plans.
- **Develop advanced monitoring systems**
  Integrating technology such as automated tracking or real-time monitoring can improve the accuracy and efficiency of bird observation systems.

These recommendations can help improve bird conservation efforts, enhance data collection practices, and support sustainable environmental planning.

Conclusions

This project successfully analyzed bird observation data from forest and grassland habitats using Python, SQL, and Power BI. Through data cleaning, transformation, and exploratory analysis, meaningful patterns in bird distribution, species diversity, and environmental influence were identified.

The analysis revealed that forest areas generally support higher bird activity and biodiversity compared to grassland habitats. It also highlighted that a few species dominate the observations, while several species appear rarely and may require closer monitoring. Environmental factors such as temperature and weather conditions were found to influence bird activity, with moderate conditions showing higher observations.

The Power BI dashboard effectively visualizes these insights, allowing interactive exploration of bird activity across habitats, locations, and environmental conditions. Overall, this project demonstrates how data-driven analysis can help understand ecological patterns, support biodiversity conservation, and assist in environmental decision-making.

This project also showcases end-to-end data analysis skills, including data preprocessing, feature engineering, SQL-based analysis, and dashboard development, making it a strong foundation for real-world data analytics applications.